THE LOCAL RING OF THE FIBRE BEING NOETHERIAN COHEN-MACAULAY IS ETALE LOCAL ON SOURCE AND TARGET FOR GERMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6192-6203 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6205-6215, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The definition of etale-local-on-source-and-target for properties of germs (Descent) and its etale-on-fibres lemma, plus the henselization-CM equivalence (More on Algebra); cited inputs with the argument written in the retained span.

Source reference. Original lemma 04NB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6215–6215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NI: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6215–6215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06LM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6215–6215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property of morphisms of germs of schemes*

$$\mathcal{P}((X, x) \to (S, s)) =$$

the local ring $\mathcal{O}_{X_s,x}$ of the fibre is Noetherian and Cohen-Macaulay

is étale local on the source-and-target (Descent, Definition 04NB).

Proof. Given a diagram as in Descent, Definition 04NB we obtain an étale morphism of fibres $U'_{v'} \to U_v$ mapping u' to u, see Descent, Lemma 04NI. Thus the strict henselizations of the local rings $\mathcal{O}_{U'_{v'},u'}$ and $\mathcal{O}_{U_v,u}$ are the same. We conclude by More on Algebra, Lemma 06LM. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.